

June 28, 2026

Editor-in-Chief
Water Research

Dear Editor,

I am pleased to submit our manuscript, “Watershed-scale PFAS fate and transport across surface water and groundwater: a two-way coupled SWAT⁺/MODFLOW 6 model quantifies the legacy groundwater contribution to in-stream loads,” for consideration as a research paper in *Water Research*.

Per- and polyfluoroalkyl substances (PFAS) are now federally regulated in drinking water, and managing them turns on *where* an observed in-stream concentration originates. Groundwater discharge is the most consequential yet least resolvable of those pathways: a grab sample cannot separate groundwater-borne from surface-derived PFAS, and watershed models have carried PFAS only from the surface into the subsurface, never returning the groundwater-borne fraction to the stream — so the groundwater *contribution* to the in-stream load, the very quantity source attribution depends on, has remained outside the model. Our work resolves this missing direction with a two-way coupling that carries PFAS through both compartments in one continuous mass balance, validated against an unusually rich public record (5,383 water-table heads, hundreds of groundwater PFAS observations, streambed porewater, and a multi-analyte in-stream network) on the Rogue River, Michigan, below the Wolverine World Wide legacy source.

The central result is that two independent lines of evidence — a model-free multi-analyte fingerprint decomposition and a double-counting-aware joint surface/groundwater calibration — converge to show that legacy groundwater discharge is a first-order, spatially localized contributor to the lower-mainstem in-stream PFOS load, with a systematic global sensitivity and uncertainty analysis identifying the controls on the surface-versus-groundwater partition. Enabling this are a process-based three-phase PFAS soil model (with air–water-interface partitioning) in SWAT⁺ and PFAS groundwater transport in MODFLOW 6’s native framework, rather than a frozen RT3D coupling.

This manuscript is original, unpublished, and not under consideration elsewhere. For transparency: three *methodological* components of the broader program are reported in separate companion manuscripts under review and cited herein — the SWATGenX model-generation platform, the SWAT⁺ engine acceleration (*Geoscientific Model Development*), and a national well-lithology inventory (*Geodata and AI*) — but the present manuscript is the only one developing and validating the coupled surface-water/groundwater PFAS model and its findings. There are no competing interests, and the work received no specific external funding. All code and data are archived openly (Zenodo and a public GitHub repository).

Thank you for considering our submission; I would be glad to suggest reviewers or provide further information.

Sincerely,

Vahid Rafiei